COMPUTER NETWORKS FINAL EXAMINATION (2024/2025) – Moad B -- SOLUTION

Gadi Taubenfeld

Question 1

Assume that we change -- True
If the routing is according -- False
A router will forward -- True
The question refers -- False
A large datagram -- True
One step in forwarding -- False
ARP queries are sent -- False
A network on the Internet -- False
Suppose a packet arrives -- False
All the computers – True

Question 2

Consider the Go-Back-N – 4 (95)
Consider the distributed algorithm – 1
Suppose an ISP owns the block – 1 (8)
Consider a client wanting to – 2 (19)

Question 3

1. $(2^{64} \times 8)/ (75 \times 10^{12}) = 1{,}967{,}652.701$ sec = 22.77 days

2.

question	Answer
1	**2^{32} , which is around 4.19 Gbytes**.
2	**385.9 seconds**

3. [1 mistake -3 points. 2 mistakes -5 points. 3 mistakes -8 points.]

Question	Answer
1	12
2	41
3	28
4	24

Question 4

1. [First mistake -1 point ; each additional mistake -1/2 point (max -8 points)]

Table for B, where poisoned reverse **is used**				
Table for B	neighbors			
destinations	A	C	E	F
A	2	infinity	17	infinity
C	infinity	2	13	infinity
D	9	7	12	6
E	infinity	5	10	8
F	infinity	infinity	14	4

2. [First mistake -3 points ; each additional mistake -2 points]

Subnet	**address -- a.b.c.d/x**
A	58.37.241.0/26
B	58.37.241.64/26
C	58.37.241.128/25
A+B+C	58.37.241.0/24

3. [each mistake -2 points]

Question	**Number of ARP+IP packets** Write **ONE** number in each entry
(1)	**12**
(2)	**4**
(3)	**2**
(4)	**8**